TANZANIA HEADS OF ISLAMIC SCHOOLS COUNCIL

FORM FOUR INTER ISLAMIC MOCK EXAMINATION

032/2A **CHEMISTRY 2A**

(ACTUAL PRACTICAL)

(For Both School and Private Candidates)

Time: 2:30 Hours **Thursday, 15th September 2022 a.m.**

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Instructions

1. This paper consists of two (2) questions.
2. Answer **BOTH** questions.
3. Cellular phones and any unauthorized materials are not allowed in the examination room.
4. Non-programmed calculators are allowed.
5. Write Your **Examination Number** on every page of your answer booklet(s).
6. The following constants might be needed in your calculations

   Atomic masses: H = 1, C = 12, O = 16, Na = 23, S = 32, Cl = 35.5

   1 litre = $1dm^3$ = $1000cm^3$ = 1000mls.

1. You are provided with the following:

   Solution PP : Made by diluting 100cm^3 of 1M to 1000cm^3 of Hydrochloric acid solution.

   Solution QQ: Made by dissolving 53g of Y_2CO_3 to make a 1 litre solution.

   Indicator MO: Methyl orange.

Procedures:

1. Fill the burette with solution PP.
2. Transfer 20 or 25cm^3 of solution QQ to a conical flask using a respective pipette.
3. Titrate solution PP against solution QQ using two drops of MO-indicator.
4. Repeat the titration to obtain three more titre values and record your results in tabular form.

Questions:

(a) (i) Complete the table of the results and work out the average volume of the acid (PP) used.

(ii) _____cm^3 of PP (acid) required _____cm^3 QQ (base) for complete neutralization.

(iii) The colour change of indicator is from _____ to _____.

(b) Calculate the molarity of the acid.

(c) (i) Write the balanced equation for the reaction between PP and QQ.

(ii) Calculate the molarity of base Y_2CO_3.

(d) Calculate:

(i) The atomic mass of Y in the formula Y_2CO_3.

(ii) Identify the element Y.

(iii) Write the chemical formula of Y_2CO_3.

2. You are provided with unknown sample of simple salt named as X made up of one cation and one anion. Carry out systematic procedures as guided below to identify the cation and anion present in the salt.

EXPERIMENT TABLE:

S/N	Experiment	Observations	Inference
(a)	Appearance of sample X.		
(b)	Add small portion of dry solid sample in a dry test tube and heat gently and then strongly until no further change.		
(c)	To a small solid sample in a test tube add dilute HCl		
(d)	Dissolve small amount of solid sample in a test tube and divide it into three portions		
	(i) To one portion add dilute NaOH till excess		
	(ii) To the second portion add dilute NH_3 solution till excess		
	(iii) To the third portion add about 3 drops of dilute HNO_3 followed by 3 drops of $AgNO_3$ solution and then NH_3 solution.		
(e)	Perform a flame test on solid sample X.		

(a) Conclusion:

(i) The cation in solid sample X is _____ and anion is _____

(ii) Name of sample X is _____

(iii) The chemical formula of sample X is _____

(b) With the aid of a balanced ionic equation explain the type of reaction between sample solution X with sodium hydroxide (NaOH).

Wabillah Tawfiq

TANZANIA HEADS OF ISLAMIC SCHOOLS COUNCIL

FORM IV INTER ISLAMIC MOCK EXAMINATION

032/2A **CHEMISTRY 2A**

MARKING SCHEME

1. (a) Table of results

(i)

	Pilot	**V_1**	**V_2**	**V_3**
Final volume (cm^3)	26.00	25.00	25.00	50.00
Initial volume (cm^3)	00.00	00.00	00.00	25.00
Volume used (cm^3)	26.00	25.00	25.00	25.00

06 marks

Average volume:

Average $= \frac{(V_1 + V_2 + V_3)cm^3}{3}$

Volume of acid $= \frac{(25.00 + 25.00 + 25.00)cm^3}{3} = \frac{75cm^3}{3}$

A.V = $25cm^3$ of acid **02 marks**

(ii) **25cm³** of PP required **25cm³** of QQ for couple reaction. **01mark**

(iii) From yellow to pink **01mark**

(b) Molarity of the acid

Data given

Initial Molarity (M_1) = 1M

Initial Volume (V_1) = $100cm^3$ = 0.1L

Final Molarity (M_2) = x

Final Volume (V_2) = $1000cm^3$ = 1L 00½

From dilution law (number of moles before dilution is equal to number of moles after dilution.

$M_1V_1 = M_2V_2$

$M_2 = \frac{M_1V_1}{V_2} = \frac{1M \times 0.1L}{1L}$ **[02marks]**

M_2=0.1M

(c) (i) Balanced equation for the reaction of PP and QQ.

$$2HCl_{(aq)} + Y_2CO_{3(aq)} \rightarrow 2YCl_{(aq)} + CO_{2(g)} + H_2O_{(l)}$$ **02marks**

(ii) Molarity of the QQ (base)

Data given

Molarity of acid (Ma) = 0.1M

Volume of acid (Va) = $25cm^3$

Molarity of base (Mb) = ?

Volume of base (Vb) = $25cm^3$

Number of acids na = 2

Number of base nb = 1. 00½

From the formula $\frac{MaVa}{na} = \frac{MbVb}{nb}$ 01

$Mb = \frac{MaNanb}{naVb}$

$Mb = \frac{01M \times 25cm^3}{2 \times 25cm^3}$

Molarity of base (Mb) = 0.05M 02 Marks

(d) (i) Atomic mass of Y is the formula Y_2CO_3.

1st concentration in grams/L of Y_2CO_3.

Data given

Mass dissolved = 53g

Volume of solution = 10 litres

Conc. g/L = $\frac{53g}{10L} = 5.3g/L$ or 53g/l

Conc. g/L of Base = 5.3g/L. **02marks**

But from the formula

Molarity of base = $\frac{Conc.\frac{g}{L} of\ base}{Molar\ mass}$

$$0.05M = \frac{5.3g/L}{Mr}$$

$$Mr = \frac{5.3g/L}{0.05}$$

Molar mass of base (Mr) = 106g/mol or 1060g/mol. 01

Atomic mass of Y in the formula Y_2CO_3.

Molar mass = 2(Y) + 12 + 3 (16) = 106/mol or 1060g/mol.

2y + 12 + 48 = 106g/mol

2y + 60 = 10g/mol **04 marks**

2y = 106 – 60

$$\frac{2y}{2} = \frac{46}{2}$$

The atomic mass of y = 23g/mol. Or 200g/mol.

(i) Element y is sodium (Na) metal **[01mark]**

(ii) Chemical formula of y is y_2CO_2 = Na_2CO_3. **[01mark]**

2. Analysis of sample salt X

RESULTS

Experiment	Observations	Inference
(a) Appearance	White deliquescent crystals	Absence of transition metal elements Fe^{24}, Fe^{3} and Cu^{2+}
(b) Small amount of solid sample was put in dry test tube and heated gently then strongly	Courless gas evolved which give dense white fumes with ammonia solution	Cl^- of hydrated Ca^{2+}, Zn^{2+} may be present
(c) Dilute Hl was added into a test tube containing a small amount of solid sample	No gas evolved	SO_3^{2-}, NO_3^- or Cl^- may be present
(d) Small amount of solid sample was mode to dissolve in a test tube by adding distilled H_2O.	Soluble in distilled OR sample dissolved in distilled H_2O.	Cl- of Zn^{2+} or Ca^{2+} may be present
(e) (i) To a small sample solution, NaOH solution was added ill excess	White precipitate was formed insoluble in excess	Ca^{2+} may be present
(ii) To a small sample solution, NH_3 solution was added until excess	No precipitate was formed	Ca^{2+} may be present

(iii) <u>Confirmation of Cl</u> 3 drops of dilute HNO_3 was added into a sample solution in a test tube followed by addition of 3 drops of $AgNO_3$ solution and excess NH_3 solution.	White precipitate soluble in excess ammonia was formed	Cl^- ion was confirmed
(e) A glass rod alcamed with conc. HCl was touched to the solid sample X and heated on a direct flame	Brick-red flame was observed	Ca^{2+} was confirmed

16 marks

2. (a) (i) Cation is Ca^{2+} ion **01**

Anion is Cl^- ion **01**

(ii) Sample X is $CaCl_2$ (Calcium chloride) **01**

(iii) Chemical formula $Ca^{2+}_{(aq)} + Cl^-_{(aq)} \rightarrow CaCl_{2(aq)}$ **01**

Formula $CaCl_2$.

(b) $CaCl_{2(aq)} + 2NaOH_{(aq)} \rightarrow Ca(OH)_{2(s)} + 2NaCl_{(aq)}$ **01**

Ions

$Ca^{2+}_{(aq)} + 2Cl^-_{aq} + 2Na^{(+)}_{(aq)} + 2OH^-_{(aq)} \rightarrow Ca(OH)_{2(s)} + 2Na^{2-}_{(aq)} + 2Cl^-_{(aq)}$ **01**

<u>Cancelation of spectator ions</u>

$Ca^{2+}_{(aq)} + 2OH^-_{(aq)} \rightarrow Ca(OH)_{2(s)}$

$Ca^{2+}_{(aq)} + 2OH^-_{(aq)} \rightarrow Ca(OH)_{2(s)}$ **01**

He reaction is a precipitation reaction. **01**

Explanation:

It is a type of chemical reaction where two aqueous solutions react to produce an in solution substance which appeared to be without or of any color. **01**

Wabillah Tawfiiq